

PHONE BOOK MANAGEMENT SYSTEM

The programming and testing

IX. Programming

9.1. Technology used

- Language: Python
- Database: MySQL
- Main library:
 - *mysql-connector-python*: Connect to and execute SQL statements.
 - *rich*: Build a professional Command Line Interface using a color palette.
 - *hashlib*: Use the SHA-256 algorithm to hash user passwords before storing them in the database.

9.2 Source code structure

- *main.py*: Navigate menus and interact with users.
- *database.py*: Contains functions for querying the database.
- *validators.py*: Check the input data format (phone number, email, password).
- *models.py*: Define object classes such as User, Contact, and Category.

9.3 Database

```
CREATE DATABASE IF NOT EXISTS phonebook;  
USE phonebook;
```

```
CREATE TABLE IF NOT EXISTS users (  
    userid INT PRIMARY KEY AUTO_INCREMENT,  
    user_name VARCHAR(255) NOT NULL UNIQUE,  
    password VARCHAR(255) NOT NULL  
);
```

```
CREATE TABLE IF NOT EXISTS categories (  
    cate_id INT PRIMARY KEY AUTO_INCREMENT,  
    cate_name VARCHAR(255) NOT NULL,  
    userid INT,  
    FOREIGN KEY (userid) REFERENCES users(userid)  
);
```

```
CREATE TABLE IF NOT EXISTS contacts (  
    contact_id INT PRIMARY KEY AUTO_INCREMENT,  
    contact_name VARCHAR(255) NOT NULL,  
    phone VARCHAR(20) NOT NULL UNIQUE,  
    email VARCHAR(255),  
    address VARCHAR(255),  
    userid INT,  
    cate_id INT,  
    FOREIGN KEY (userid) REFERENCES users(userid),  
    FOREIGN KEY (cate_id) REFERENCES categories(cate_id)  
);
```

9.4 Important processing logic

9.4.1 Security Logic

- The system uses the *hashlib* library to hash passwords using the SHA-256 algorithm. This ensures absolute security; passwords stored in the database are unique and cannot be reversed.

9.4.2 Validation Logic

- Use *Regular Expression* in the *validators.py* file to control input data. The system only accepts phone numbers in the standard Vietnamese format (10 digits, starting with 0 or +84).

9.4.3 Database Integration

- Use queries with *%s* parameters to connect Python to MySQL, effectively preventing SQL Injection vulnerabilities.

X. System testing

10.1 Test case

The following tables present the complete set of test cases designed to verify all functional requirements of the Phone Book Management System. A total of 21 test cases were prepared, covering authentication, contact management, search & filter, contact operations, and category & session management.

10.1.1 Authentication Testing

Test Case ID	Function	Test Steps	Input Data	Expected Result
TC01	Register	Enter valid information	Username: student01 Password: student123	The message "Account has been successfully registered"
TC02	Register	Enter a username with fewer than 3 characters	Username: st	Error message: "Invalid username! Must be 3 or more characters (a-z, 0-9, _)"
TC03	Register	Enter a password with at least 6 characters	Password: 12345	Error message: "Passwords must be at least 6 characters long!"
TC04	Login	Enter the correct username/password	Username: student01 Password: student123	Login successful, now go to the main menu
TC05	Login	Incorrect password entered	Username: student01 Password: wrongpass	Error message: "Incorrect username or password"

10.1.2 Contact Management Testing

Test Case ID	Function	Test Steps	Input Data	Expected Result
TC06	Add Contact	Enter your phone number in the correct Vietnamese format	Name: Ngan Phone: 0964634548 or +84964634548	Saved successfully, notification "Contact added successfully"
TC07	Add Contact	Entered phone number in the wrong format	Phone: 12345 or +123...	Error message: "Invalid phone number"

TC08	Add Contact	Enter a phone number that is already in your contacts	Phone: 0964634548 (already exists)	Error message: "This phone number already exists!"
TC09	Add Contact	Leave Email blank (Optional)	Email: [Leave blank]	The system still allows saving successfully

10.1.3 Search & Filter Testing

Test Case ID	Function	Test Steps	Input Data	Expected Result
TC10	Filter	1. Select function 6 2. Enter the category ID you want to filter	ID: 1 (Example: Friend)	The system displays a list of contacts belonging to the category with the corresponding ID that has been selected
TC11	Filter	1. Select function 6 2. Enter an ID that has no data yet	ID: 12	The system displays the message: "No contacts found in this category!" (or an empty table)

10.1.4 Contact Operations

Test Case ID	Function	Test Steps	Input Data	Expected Result
TC12	Edit Contact	1. Select function 2 2. Enter contact ID 3. Update email	Contact ID:1 New Email:nhi@gmail.com	Contact is updated successfully
TC13	Edit Contact	1. Select function 2 2. Enter contact ID 3. Change category only	Contact ID:2 Category: Friend	Contact category updated successfully
TC14	Delete Contact	1. Select function 3 2. Enter contact ID 3. Confirm delete	Contact ID:1	Contact is deleted successfully
TC15	Search Contact	1. Select function 4 2. Enter contact name	Name: Nhi	Matching contact is displayed
TC16	Search Contact	1. Select function 4 2. Enter phone number	Phone:0964634548	Matching contact displayed
TC17	Search Contact	Search non-existing contact	Name: XYZ	No contact found
TC18	Display All Contacts	Select function 5	None	All contacts are displayed

10.1.5 Category & Session

Test Case ID	Function	Test Steps	Input Data	Expected Result
TC19	Add Category	1. Select function 7 2. Select submenu 1 3. Enter category name	Category: Family	Category added successfully
TC20	Delete Category	1. Select function 7 2. Select submenu 2 3. Enter category ID 4. Confirm y	Category ID:1	Category deleted successfully
TC21	Logout	Select function 0	None	User logs out successfully and returns to authentication menu

10.2 Debugging

During execution of all defined test cases (TC01–TC21), no defects were detected. All functionalities operated as expected, including authentication, contact management, search and filter, category management, and logout.

No discrepancies were found between expected results and actual results.

Therefore:

- No debugging fixes were required.
- No source code modifications were needed after test execution.
- All test cases passed successfully during initial testing.

Debugging Result:

- No defects were detected during testing.
- No debugging actions were required.

10.3 Bug Management

No defects were identified during the execution of all test cases. Therefore, no bug tickets were created and no bug tracking workflow was required.

The bug management process concluded with:

- Bugs Reported: 0
- Bugs Resolved: 0
- Open Bugs: 0

Bug Status:

- No bugs recorded.
- No bug management actions required.

10.4 Testing Summary

Total Test Cases:21

Passed:21

Failed:0

Detected Bugs:0

Resolved Bugs:0

Final Pass Rate:100%